# Retail Sales Database Analysis

SQL project | SQLite | Sales of 2025

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This project builds a small relational sales database for a retail company (customers, products, orders and order items) and answers business questions with SQL: JOINs, GROUP BY, CTEs and window functions. All money values are in Rs. The data is sample data created for this project.

## Database tables

| Table | Columns | Rows |
|---|---|---|
| customers | customer_id, name, city, signup_date | 40 |
| products | product_id, name, category, price, cost | 15 |
| orders | order_id, customer_id, order_date | 240 |
| order_items | item_id, order_id, product_id, quantity, discount | 384 |

## Key findings

• Total revenue was Rs 22,578,865 from 240 orders placed by 32 customers. Profit was Rs 5,423,065 (margin 24.0%) and the average order value was Rs 94,079.

• The best month was 2025-12 (Rs 3,883,498) and the weakest was 2025-05 (Rs 627,040). Revenue grew compared with the previous month in 6 of 11 months. The year ends with Rs 22,578,865 in total.

• Laptop is the top product with Rs 6,354,750 revenue (36 units). The top 5 products together bring Rs 18,854,750.

• Electronics earns the most revenue (Rs 12,998,800), while Stationery has the highest profit margin (39.8%).

• Faisalabad ranks first (Rs 7,994,880) and Lahore ranks last (Rs 659,778).

• The top 10 customers produce 71.0% of total revenue. The biggest customer is Maryam Siddiqui (Faisalabad) with Rs 3,301,342.

• 8 registered customers have never ordered. They are good targets for a first-purchase offer.

# Analysis queries

## Q1. Business summary (KPIs)

Total orders, customers, revenue, profit and average order value. Revenue = quantity x price x (1 - discount); profit subtracts the product cost.

```
SELECT COUNT(DISTINCT o.order_id)    AS total_orders,
       COUNT(DISTINCT o.customer_id) AS active_customers,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount)), 2) AS total_revenue,
       ROUND(SUM(oi.quantity * (p.price * (1 - oi.discount) - p.cost)), 2) AS total_profit,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount))
             / COUNT(DISTINCT o.order_id), 2) AS avg_order_value
FROM orders o
JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p     ON p.product_id = oi.product_id;
```

| Total orders | Active customers | Total revenue | Total profit | Avg order value |
|---|---|---|---|---|
| 240 | 32 | 22,578,865 | 5,423,065 | 94,078.60 |

**Insight:** Total revenue was Rs 22,578,865 from 240 orders placed by 32 customers. Profit was Rs 5,423,065 (margin 24.0%) and the average order value was Rs 94,079.

## Q2. Monthly revenue, growth and running total

Uses a CTE plus the window functions LAG (previous month) and SUM() OVER (running total).

```
WITH monthly AS (
    SELECT strftime('%Y-%m', o.order_date) AS month,
           SUM(oi.quantity * p.price * (1 - oi.discount)) AS revenue
    FROM orders o
    JOIN order_items oi ON oi.order_id = o.order_id
    JOIN products p     ON p.product_id = oi.product_id
    GROUP BY month
)
SELECT month,
       ROUND(revenue, 2) AS revenue,
       ROUND((revenue - LAG(revenue) OVER (ORDER BY month)) * 100.0
             / LAG(revenue) OVER (ORDER BY month), 1) AS growth_pct,
       ROUND(SUM(revenue) OVER (ORDER BY month), 2) AS running_total
FROM monthly
ORDER BY month;
```

| Month | Revenue | Growth pct | Running total |
|---|---|---|---|
| 2025-01 | 2,043,740 | - | 2,043,740 |
| 2025-02 | 1,555,130 | -23.9 | 3,598,870 |
| 2025-03 | 2,245,757.50 | 44.4 | 5,844,627.50 |
| 2025-04 | 2,851,107.50 | 27.0 | 8,695,735 |
| 2025-05 | 627,040 | -78.0 | 9,322,775 |
| 2025-06 | 1,317,257.50 | 110.1 | 10,640,032.50 |
| 2025-07 | 1,270,395 | -3.6 | 11,910,427.50 |
| 2025-08 | 912,722.50 | -28.2 | 12,823,150 |
| 2025-09 | 3,526,265 | 286.3 | 16,349,415 |
| 2025-10 | 982,960 | -72.1 | 17,332,375 |
| 2025-11 | 1,362,992.50 | 38.7 | 18,695,367.50 |
| 2025-12 | 3,883,497.50 | 184.9 | 22,578,865 |

**Insight:** The best month was 2025-12 (Rs 3,883,498) and the weakest was 2025-05 (Rs 627,040). Revenue grew compared with the previous month in 6 of 11 months. The year ends with Rs 22,578,865 in total.

## Q3. Top 5 products by revenue

JOIN + GROUP BY + ORDER BY + LIMIT.

```
SELECT p.name AS product,
       p.category,
       SUM(oi.quantity) AS units_sold,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount)), 2) AS revenue
FROM order_items oi
JOIN products p ON p.product_id = oi.product_id
GROUP BY p.product_id
ORDER BY revenue DESC
LIMIT 5;
```

| Product | Category | Units sold | Revenue |
|---|---|---|---|
| Laptop | Electronics | 36 | 6,354,750 |
| Sofa Set | Furniture | 48 | 5,430,000 |
| Smartphone | Electronics | 36 | 3,310,750 |
| Study Desk | Furniture | 56 | 1,881,250 |
| Tablet | Electronics | 32 | 1,878,000 |

**Insight:** Laptop is the top product with Rs 6,354,750 revenue (36 units). The top 5 products together bring Rs 18,854,750.

## Q4. Revenue, profit and margin by category

Which category earns the most, and which earns the best margin?

```
SELECT p.category,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount)), 2) AS revenue,
       ROUND(SUM(oi.quantity * (p.price * (1 - oi.discount) - p.cost)), 2) AS profit,
       ROUND(100.0 * SUM(oi.quantity * (p.price * (1 - oi.discount) - p.cost))
             / SUM(oi.quantity * p.price * (1 - oi.discount)), 1) AS margin_pct
FROM order_items oi
JOIN products p ON p.product_id = oi.product_id
GROUP BY p.category
ORDER BY revenue DESC;
```

| Category | Revenue | Profit | Margin pct |
|---|---|---|---|
| Electronics | 12,998,800 | 2,250,800 | 17.3 |
| Furniture | 9,172,050 | 3,010,050 | 32.8 |
| Stationery | 408,015 | 162,215 | 39.8 |

**Insight:** Electronics earns the most revenue (Rs 12,998,800), while Stationery has the highest profit margin (39.8%).

## Q5. Cities ranked by revenue

Joins four tables and uses RANK() OVER to rank cities.

```
SELECT c.city,
       COUNT(DISTINCT o.order_id) AS orders,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount)), 2) AS revenue,
       RANK() OVER (ORDER BY SUM(oi.quantity * p.price * (1 - oi.discount)) DESC) AS city_rank
FROM customers c
JOIN orders o       ON o.customer_id = c.customer_id
JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p     ON p.product_id = oi.product_id
GROUP BY c.city
ORDER BY city_rank;
```

| City | Orders | Revenue | City rank |
|---|---|---|---|
| Faisalabad | 79 | 7,994,880 | 1 |
| Karachi | 66 | 6,331,125 | 2 |
| Multan | 48 | 4,038,862.50 | 3 |
| Islamabad | 37 | 3,554,220 | 4 |
| Lahore | 10 | 659,777.50 | 5 |

**Insight:** Faisalabad ranks first (Rs 7,994,880) and Lahore ranks last (Rs 659,778).

## Q6. Top 10 customers and their share of revenue

A CTE for customer totals, then a subquery for the overall total to get the share.

```
WITH customer_sales AS (
    SELECT c.customer_id, c.name, c.city,
           SUM(oi.quantity * p.price * (1 - oi.discount)) AS revenue
    FROM customers c
    JOIN orders o       ON o.customer_id = c.customer_id
    JOIN order_items oi ON oi.order_id = o.order_id
    JOIN products p     ON p.product_id = oi.product_id
    GROUP BY c.customer_id
)
SELECT name, city,
       ROUND(revenue, 2) AS revenue,
       ROUND(100.0 * revenue / (SELECT SUM(revenue) FROM customer_sales), 1) AS share_pct
FROM customer_sales
ORDER BY revenue DESC
LIMIT 10;
```

| Name | City | Revenue | Share pct |
|---|---|---|---|
| Maryam Siddiqui | Faisalabad | 3,301,342.50 | 14.6 |
| Ayesha Chaudhry | Islamabad | 2,729,530 | 12.1 |
| Hassan Butt | Faisalabad | 1,620,280 | 7.2 |
| Hassan Khan | Karachi | 1,497,892.50 | 6.6 |
| Rabia Qureshi | Karachi | 1,454,320 | 6.4 |
| Farhan Butt | Karachi | 1,390,700 | 6.2 |
| Kiran Ansari | Multan | 1,372,122.50 | 6.1 |
| Fatima Malik | Faisalabad | 959,495 | 4.2 |
| Rabia Hussain | Faisalabad | 864,990 | 3.8 |
| Rabia Raza | Multan | 849,690 | 3.8 |

**Insight:** The top 10 customers produce 71.0% of total revenue. The biggest customer is Maryam Siddiqui (Faisalabad) with Rs 3,301,342.

## Q7. Customers who never placed an order

LEFT JOIN with IS NULL finds rows with no match in the other table.

```
SELECT c.customer_id, c.name, c.city, c.signup_date
FROM customers c
LEFT JOIN orders o ON o.customer_id = c.customer_id
WHERE o.order_id IS NULL;
```

| Customer id | Name | City | Signup date |
|---|---|---|---|
| 4 | Hassan Raza | Faisalabad | 2024-07-26 |
| 27 | Hamza Ansari | Multan | 2024-04-27 |
| 35 | Hamza Malik | Lahore | 2024-01-11 |
| 36 | Saad Hussain | Multan | 2024-06-24 |
| 37 | Ali Qureshi | Faisalabad | 2025-03-12 |
| 38 | Zainab Hussain | Lahore | 2024-10-22 |
| 39 | Kiran Butt | Multan | 2025-05-26 |
| 40 | Omar Hussain | Lahore | 2024-05-10 |

**Insight:** 8 registered customers have never ordered. They are good targets for a first-purchase offer.

# Appendix A: Sales list

First 45 records out of 384 orders lines, joined with customers and products. Query used:

```
SELECT o.order_id, o.order_date, c.name AS customer, c.city,
       p.name AS product, oi.quantity,
       ROUND(oi.discount * 100) AS discount_pct,
       ROUND(oi.quantity * p.price * (1 - oi.discount), 2) AS amount
FROM orders o
JOIN customers c    ON c.customer_id = o.customer_id
JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p     ON p.product_id = oi.product_id
ORDER BY o.order_date, o.order_id, oi.item_id
LIMIT 45;
```

| Order id | Order date | Customer | City | Product | Quantity | Discount pct | Amount |
|---|---|---|---|---|---|---|---|
| 1 | 2025-01-01 | Maryam Siddiqui | Faisalabad | Office Chair | 1 | 10.0 | 22,500 |
| 1 | 2025-01-01 | Maryam Siddiqui | Faisalabad | Sofa Set | 2 | 0.0 | 240,000 |
| 2 | 2025-01-03 | Ayesha Chaudhry | Islamabad | Laptop | 2 | 0.0 | 370,000 |
| 3 | 2025-01-08 | Rabia Raza | Multan | File Folder | 4 | 0.0 | 1,400 |
| 3 | 2025-01-08 | Rabia Raza | Multan | Smartphone | 2 | 5.0 | 180,500 |
| 4 | 2025-01-09 | Kiran Ansari | Multan | Pen Set | 4 | 0.0 | 1,800 |
| 5 | 2025-01-16 | Hassan Malik | Karachi | Tablet | 2 | 10.0 | 108,000 |
| 6 | 2025-01-17 | Ayesha Chaudhry | Islamabad | Printer Paper | 2 | 0.0 | 3,600 |
| 6 | 2025-01-17 | Ayesha Chaudhry | Islamabad | Calculator | 3 | 10.0 | 5,940 |
| 7 | 2025-01-18 | Maryam Siddiqui | Faisalabad | Study Desk | 2 | 0.0 | 70,000 |
| 7 | 2025-01-18 | Maryam Siddiqui | Faisalabad | Sofa Set | 2 | 10.0 | 216,000 |
| 8 | 2025-01-20 | Maryam Siddiqui | Faisalabad | Bookshelf | 3 | 5.0 | 51,300 |
| 9 | 2025-01-21 | Hassan Sheikh | Faisalabad | Calculator | 4 | 0.0 | 8,800 |
| 10 | 2025-01-23 | Zainab Chaudhry | Karachi | Marker Set | 4 | 20.0 | 2,880 |
| 11 | 2025-01-23 | Noor Khan | Islamabad | Calculator | 2 | 0.0 | 4,400 |
| 11 | 2025-01-23 | Noor Khan | Islamabad | Study Desk | 2 | 10.0 | 63,000 |
| 12 | 2025-01-23 | Fatima Malik | Faisalabad | Smart Watch | 2 | 0.0 | 44,000 |
| 12 | 2025-01-23 | Fatima Malik | Faisalabad | Laptop | 1 | 0.0 | 185,000 |
| 13 | 2025-01-26 | Saad Chaudhry | Karachi | Smart Watch | 2 | 20.0 | 35,200 |
| 13 | 2025-01-26 | Saad Chaudhry | Karachi | Calculator | 4 | 10.0 | 7,920 |
| 13 | 2025-01-26 | Saad Chaudhry | Karachi | Smartphone | 2 | 20.0 | 152,000 |
| 14 | 2025-01-27 | Hassan Butt | Faisalabad | Smartphone | 2 | 5.0 | 180,500 |
| 15 | 2025-01-28 | Hira Malik | Multan | Headphones | 2 | 0.0 | 17,000 |
| 15 | 2025-01-28 | Hira Malik | Multan | Bookshelf | 4 | 0.0 | 72,000 |
| 16 | 2025-02-04 | Hassan Khan | Karachi | Printer Paper | 2 | 0.0 | 3,600 |
| 16 | 2025-02-04 | Hassan Khan | Karachi | Laptop | 2 | 0.0 | 370,000 |
| 16 | 2025-02-04 | Hassan Khan | Karachi | Bookshelf | 1 | 0.0 | 18,000 |
| 17 | 2025-02-05 | Hira Malik | Multan | Printer Paper | 1 | 20.0 | 1,440 |
| 18 | 2025-02-05 | Fatima Malik | Faisalabad | Laptop | 1 | 20.0 | 148,000 |
| 18 | 2025-02-05 | Fatima Malik | Faisalabad | File Folder | 2 | 0.0 | 700 |

| Order id | Order date | Customer | City | Product | Quantity | Discount pct | Amount |
|---|---|---|---|---|---|---|---|
| 18 | 2025-02-05 | Fatima Malik | Faisalabad | Printer Paper | 1 | 0.0 | 1,800 |
| 19 | 2025-02-09 | Ayesha Chaudhry | Islamabad | Notebook Pack | 3 | 10.0 | 1,620 |
| 20 | 2025-02-10 | Farhan Butt | Karachi | Office Chair | 2 | 10.0 | 45,000 |
| 21 | 2025-02-11 | Hassan Khan | Karachi | Pen Set | 1 | 0.0 | 450 |
| 22 | 2025-02-12 | Saad Malik | Islamabad | Smartphone | 1 | 0.0 | 95,000 |
| 22 | 2025-02-12 | Saad Malik | Islamabad | File Folder | 4 | 0.0 | 1,400 |
| 23 | 2025-02-12 | Kiran Malik | Faisalabad | File Folder | 3 | 20.0 | 840 |
| 24 | 2025-02-15 | Kiran Malik | Faisalabad | Smartphone | 2 | 0.0 | 190,000 |
| 25 | 2025-02-18 | Zainab Qureshi | Multan | Bookshelf | 3 | 0.0 | 54,000 |
| 26 | 2025-02-19 | Hira Malik | Multan | File Folder | 3 | 0.0 | 1,050 |
| 27 | 2025-02-19 | Rabia Siddiqui | Karachi | Office Chair | 2 | 0.0 | 50,000 |
| 28 | 2025-02-20 | Ayesha Chaudhry | Islamabad | Headphones | 4 | 5.0 | 32,300 |
| 28 | 2025-02-20 | Ayesha Chaudhry | Islamabad | Calculator | 3 | 5.0 | 6,270 |
| 29 | 2025-02-22 | Maryam Siddiqui | Faisalabad | Sofa Set | 1 | 0.0 | 120,000 |
| 30 | 2025-02-22 | Hassan Butt | Faisalabad | Tablet | 2 | 10.0 | 108,000 |

# Appendix B: Full SQL script

Create tables, insert data and run all queries in one file.

```
-- Retail Sales Database Analysis
-- Run in SQLite (sqlite3 or DB Browser for SQLite)

-- PART 1: CREATE TABLES
CREATE TABLE customers (
    customer_id INTEGER PRIMARY KEY,
    name        TEXT,
    city        TEXT,
    signup_date TEXT
);
CREATE TABLE products (
    product_id  INTEGER PRIMARY KEY,
    name        TEXT,
    category    TEXT,
    price       INTEGER,   -- selling price (Rs)
    cost        INTEGER    -- purchase cost (Rs)
);
CREATE TABLE orders (
    order_id    INTEGER PRIMARY KEY,
    customer_id INTEGER REFERENCES customers(customer_id),
    order_date  TEXT       -- YYYY-MM-DD
);
CREATE TABLE order_items (
    item_id     INTEGER PRIMARY KEY,
    order_id    INTEGER REFERENCES orders(order_id),
    product_id  INTEGER REFERENCES products(product_id),
    quantity    INTEGER,
    discount    REAL       -- 0.10 means 10% discount
);

-- PART 2: INSERT DATA
INSERT INTO customers VALUES (1, 'Farhan Chaudhry', 'Faisalabad', '2024-05-10');
INSERT INTO customers VALUES (2, 'Farhan Butt', 'Karachi', '2024-01-20');
INSERT INTO customers VALUES (3, 'Noor Khan', 'Islamabad', '2024-04-30');
INSERT INTO customers VALUES (4, 'Hassan Raza', 'Faisalabad', '2024-07-26');
INSERT INTO customers VALUES (5, 'Asad Raza', 'Lahore', '2024-10-08');
INSERT INTO customers VALUES (6, 'Hassan Butt', 'Faisalabad', '2025-06-27');
INSERT INTO customers VALUES (7, 'Ayesha Butt', 'Faisalabad', '2025-02-22');
INSERT INTO customers VALUES (8, 'Rabia Raza', 'Multan', '2024-07-22');
INSERT INTO customers VALUES (9, 'Asad Qureshi', 'Lahore', '2024-01-11');
INSERT INTO customers VALUES (10, 'Rabia Qureshi', 'Karachi', '2024-04-05');
INSERT INTO customers VALUES (11, 'Noor Siddiqui', 'Karachi', '2025-08-20');
INSERT INTO customers VALUES (12, 'Hira Malik', 'Multan', '2024-10-02');
INSERT INTO customers VALUES (13, 'Omar Khan', 'Multan', '2025-02-18');
INSERT INTO customers VALUES (14, 'Ahmed Sheikh', 'Islamabad', '2024-02-02');
INSERT INTO customers VALUES (15, 'Hassan Khan', 'Karachi', '2024-01-21');
INSERT INTO customers VALUES (16, 'Hassan Malik', 'Karachi', '2025-05-22');
INSERT INTO customers VALUES (17, 'Zainab Chaudhry', 'Karachi', '2024-12-18');
INSERT INTO customers VALUES (18, 'Zainab Qureshi', 'Multan', '2024-07-04');
INSERT INTO customers VALUES (19, 'Hira Sheikh', 'Lahore', '2024-07-09');
INSERT INTO customers VALUES (20, 'Maryam Siddiqui', 'Faisalabad', '2024-02-23');
INSERT INTO customers VALUES (21, 'Maryam Hussain', 'Islamabad', '2024-04-13');
INSERT INTO customers VALUES (22, 'Rabia Siddiqui', 'Karachi', '2024-01-15');
INSERT INTO customers VALUES (23, 'Hira Hussain', 'Karachi', '2025-05-18');
INSERT INTO customers VALUES (24, 'Ayesha Chaudhry', 'Islamabad', '2024-10-06');
INSERT INTO customers VALUES (25, 'Rabia Hussain', 'Faisalabad', '2024-03-17');
INSERT INTO customers VALUES (26, 'Bilal Chaudhry', 'Multan', '2025-03-19');
INSERT INTO customers VALUES (27, 'Hamza Ansari', 'Multan', '2024-04-27');
INSERT INTO customers VALUES (28, 'Saad Chaudhry', 'Karachi', '2024-02-07');
INSERT INTO customers VALUES (29, 'Kiran Ansari', 'Multan', '2024-12-25');
INSERT INTO customers VALUES (30, 'Fatima Malik', 'Faisalabad', '2025-06-21');
INSERT INTO customers VALUES (31, 'Ayesha Ansari', 'Islamabad', '2024-01-17');
INSERT INTO customers VALUES (32, 'Saad Malik', 'Islamabad', '2025-04-28');
INSERT INTO customers VALUES (33, 'Hassan Sheikh', 'Faisalabad', '2024-04-24');
INSERT INTO customers VALUES (34, 'Kiran Malik', 'Faisalabad', '2024-02-25');
INSERT INTO customers VALUES (35, 'Hamza Malik', 'Lahore', '2024-01-11');
INSERT INTO customers VALUES (36, 'Saad Hussain', 'Multan', '2024-06-24');
INSERT INTO customers VALUES (37, 'Ali Qureshi', 'Faisalabad', '2025-03-12');
INSERT INTO customers VALUES (38, 'Zainab Hussain', 'Lahore', '2024-10-22');
INSERT INTO customers VALUES (39, 'Kiran Butt', 'Multan', '2025-05-26');
INSERT INTO customers VALUES (40, 'Omar Hussain', 'Lahore', '2024-05-10');
INSERT INTO products VALUES (1, 'Laptop', 'Electronics', 185000, 150000);
INSERT INTO products VALUES (2, 'Smartphone', 'Electronics', 95000, 78000);
INSERT INTO products VALUES (3, 'Headphones', 'Electronics', 8500, 5500);
INSERT INTO products VALUES (4, 'Smart Watch', 'Electronics', 22000, 15000);
INSERT INTO products VALUES (5, 'Tablet', 'Electronics', 60000, 48000);
INSERT INTO products VALUES (6, 'Office Chair', 'Furniture', 25000, 15000);
INSERT INTO products VALUES (7, 'Study Desk', 'Furniture', 35000, 21000);
INSERT INTO products VALUES (8, 'Bookshelf', 'Furniture', 18000, 10500);
INSERT INTO products VALUES (9, 'Sofa Set', 'Furniture', 120000, 80000);
INSERT INTO products VALUES (10, 'Notebook Pack', 'Stationery', 600, 300);
INSERT INTO products VALUES (11, 'Pen Set', 'Stationery', 450, 200);
```

```
INSERT INTO products VALUES (12, 'Printer Paper', 'Stationery', 1800, 1200);
INSERT INTO products VALUES (13, 'Marker Set', 'Stationery', 900, 450);
INSERT INTO products VALUES (14, 'File Folder', 'Stationery', 350, 150);
INSERT INTO products VALUES (15, 'Calculator', 'Stationery', 2200, 1300);
INSERT INTO orders VALUES (1, 20, '2025-01-01');
INSERT INTO orders VALUES (2, 24, '2025-01-03');
INSERT INTO orders VALUES (3, 8, '2025-01-08');
INSERT INTO orders VALUES (4, 29, '2025-01-09');
INSERT INTO orders VALUES (5, 16, '2025-01-16');
INSERT INTO orders VALUES (6, 24, '2025-01-17');
INSERT INTO orders VALUES (7, 20, '2025-01-18');
INSERT INTO orders VALUES (8, 20, '2025-01-20');
INSERT INTO orders VALUES (9, 33, '2025-01-21');
INSERT INTO orders VALUES (10, 17, '2025-01-23');
INSERT INTO orders VALUES (11, 3, '2025-01-23');
INSERT INTO orders VALUES (12, 30, '2025-01-23');
INSERT INTO orders VALUES (13, 28, '2025-01-26');
INSERT INTO orders VALUES (14, 6, '2025-01-27');
INSERT INTO orders VALUES (15, 12, '2025-01-28');
INSERT INTO orders VALUES (16, 15, '2025-02-04');
INSERT INTO orders VALUES (17, 12, '2025-02-05');
INSERT INTO orders VALUES (18, 30, '2025-02-05');
INSERT INTO orders VALUES (19, 24, '2025-02-09');
INSERT INTO orders VALUES (20, 2, '2025-02-10');
INSERT INTO orders VALUES (21, 15, '2025-02-11');
INSERT INTO orders VALUES (22, 32, '2025-02-12');
INSERT INTO orders VALUES (23, 34, '2025-02-12');
INSERT INTO orders VALUES (24, 34, '2025-02-15');
INSERT INTO orders VALUES (25, 18, '2025-02-18');
INSERT INTO orders VALUES (26, 12, '2025-02-19');
INSERT INTO orders VALUES (27, 22, '2025-02-19');
INSERT INTO orders VALUES (28, 24, '2025-02-20');
INSERT INTO orders VALUES (29, 20, '2025-02-22');
INSERT INTO orders VALUES (30, 6, '2025-02-22');
INSERT INTO orders VALUES (31, 20, '2025-02-23');
INSERT INTO orders VALUES (32, 22, '2025-02-24');
INSERT INTO orders VALUES (33, 2, '2025-02-24');
INSERT INTO orders VALUES (34, 10, '2025-02-26');
INSERT INTO orders VALUES (35, 21, '2025-02-28');
INSERT INTO orders VALUES (36, 12, '2025-03-01');
INSERT INTO orders VALUES (37, 12, '2025-03-01');
INSERT INTO orders VALUES (38, 20, '2025-03-01');
INSERT INTO orders VALUES (39, 29, '2025-03-04');
INSERT INTO orders VALUES (40, 9, '2025-03-05');
INSERT INTO orders VALUES (41, 9, '2025-03-05');
INSERT INTO orders VALUES (42, 32, '2025-03-07');
INSERT INTO orders VALUES (43, 34, '2025-03-12');
INSERT INTO orders VALUES (44, 10, '2025-03-14');
INSERT INTO orders VALUES (45, 2, '2025-03-15');
INSERT INTO orders VALUES (46, 13, '2025-03-16');
INSERT INTO orders VALUES (47, 12, '2025-03-16');
INSERT INTO orders VALUES (48, 9, '2025-03-17');
INSERT INTO orders VALUES (49, 10, '2025-03-18');
INSERT INTO orders VALUES (50, 32, '2025-03-20');
INSERT INTO orders VALUES (51, 24, '2025-03-20');
INSERT INTO orders VALUES (52, 2, '2025-03-21');
INSERT INTO orders VALUES (53, 15, '2025-03-21');
INSERT INTO orders VALUES (54, 2, '2025-03-24');
INSERT INTO orders VALUES (55, 25, '2025-03-25');
INSERT INTO orders VALUES (56, 11, '2025-03-27');
INSERT INTO orders VALUES (57, 33, '2025-03-27');
INSERT INTO orders VALUES (58, 1, '2025-03-27');
INSERT INTO orders VALUES (59, 15, '2025-03-28');
INSERT INTO orders VALUES (60, 25, '2025-04-01');
INSERT INTO orders VALUES (61, 7, '2025-04-01');
INSERT INTO orders VALUES (62, 2, '2025-04-02');
INSERT INTO orders VALUES (63, 6, '2025-04-02');
INSERT INTO orders VALUES (64, 1, '2025-04-05');
INSERT INTO orders VALUES (65, 25, '2025-04-09');
INSERT INTO orders VALUES (66, 18, '2025-04-10');
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INSERT INTO order_items VALUES (20, 13, 15, 4, 0.1);
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INSERT INTO order_items VALUES (230, 143, 11, 4, 0.2);
INSERT INTO order_items VALUES (231, 144, 9, 2, 0.0);
INSERT INTO order_items VALUES (232, 144, 2, 1, 0.0);
INSERT INTO order_items VALUES (233, 145, 14, 1, 0.0);
INSERT INTO order_items VALUES (234, 145, 4, 1, 0.05);
INSERT INTO order_items VALUES (235, 146, 9, 1, 0.0);
INSERT INTO order_items VALUES (236, 147, 8, 4, 0.0);
INSERT INTO order_items VALUES (237, 147, 2, 1, 0.0);
INSERT INTO order_items VALUES (238, 148, 10, 1, 0.0);
INSERT INTO order_items VALUES (239, 148, 2, 1, 0.05);
INSERT INTO order_items VALUES (240, 148, 4, 2, 0.0);
INSERT INTO order_items VALUES (241, 149, 7, 1, 0.0);
INSERT INTO order_items VALUES (242, 149, 15, 2, 0.0);
INSERT INTO order_items VALUES (243, 150, 2, 2, 0.0);
INSERT INTO order_items VALUES (244, 151, 6, 2, 0.0);
INSERT INTO order_items VALUES (245, 151, 10, 1, 0.2);
INSERT INTO order_items VALUES (246, 152, 13, 2, 0.05);
INSERT INTO order_items VALUES (247, 152, 15, 2, 0.0);
INSERT INTO order_items VALUES (248, 153, 9, 2, 0.2);
INSERT INTO order_items VALUES (249, 153, 1, 1, 0.0);
INSERT INTO order_items VALUES (250, 154, 10, 3, 0.05);
INSERT INTO order_items VALUES (251, 154, 1, 2, 0.05);
INSERT INTO order_items VALUES (252, 155, 9, 1, 0.1);
INSERT INTO order_items VALUES (253, 155, 7, 1, 0.05);
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INSERT INTO order_items VALUES (255, 157, 11, 1, 0.1);
INSERT INTO order_items VALUES (256, 158, 4, 2, 0.2);
INSERT INTO order_items VALUES (257, 158, 2, 1, 0.0);
INSERT INTO order_items VALUES (258, 158, 5, 2, 0.0);
INSERT INTO order_items VALUES (259, 159, 10, 4, 0.0);
INSERT INTO order_items VALUES (260, 160, 6, 2, 0.0);
INSERT INTO order_items VALUES (261, 160, 15, 4, 0.0);
INSERT INTO order_items VALUES (262, 161, 1, 1, 0.0);
INSERT INTO order_items VALUES (263, 162, 4, 1, 0.0);
INSERT INTO order_items VALUES (264, 162, 2, 1, 0.0);
INSERT INTO order_items VALUES (265, 162, 12, 1, 0.0);
INSERT INTO order_items VALUES (266, 163, 7, 2, 0.0);
INSERT INTO order_items VALUES (267, 163, 13, 3, 0.1);
INSERT INTO order_items VALUES (268, 163, 14, 4, 0.05);
INSERT INTO order_items VALUES (269, 164, 3, 4, 0.05);
INSERT INTO order_items VALUES (270, 164, 11, 1, 0.0);
INSERT INTO order_items VALUES (271, 165, 12, 3, 0.2);
INSERT INTO order_items VALUES (272, 165, 2, 2, 0.0);
INSERT INTO order_items VALUES (273, 165, 9, 2, 0.0);
INSERT INTO order_items VALUES (274, 166, 6, 1, 0.0);
INSERT INTO order_items VALUES (275, 167, 9, 2, 0.05);
INSERT INTO order_items VALUES (276, 167, 3, 4, 0.05);
INSERT INTO order_items VALUES (277, 167, 12, 3, 0.0);
INSERT INTO order_items VALUES (278, 168, 15, 2, 0.05);
INSERT INTO order_items VALUES (279, 169, 15, 4, 0.0);
INSERT INTO order_items VALUES (280, 170, 3, 2, 0.2);
INSERT INTO order_items VALUES (281, 171, 2, 1, 0.0);
INSERT INTO order_items VALUES (282, 172, 2, 1, 0.0);
INSERT INTO order_items VALUES (283, 173, 5, 1, 0.0);
INSERT INTO order_items VALUES (284, 174, 3, 4, 0.05);
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INSERT INTO order_items VALUES (285, 175, 7, 1, 0.1);
INSERT INTO order_items VALUES (286, 175, 9, 1, 0.0);
INSERT INTO order_items VALUES (287, 176, 15, 1, 0.05);
INSERT INTO order_items VALUES (288, 177, 10, 1, 0.05);
INSERT INTO order_items VALUES (289, 178, 9, 2, 0.0);
INSERT INTO order_items VALUES (290, 178, 2, 2, 0.05);
INSERT INTO order_items VALUES (291, 179, 6, 2, 0.0);
INSERT INTO order_items VALUES (292, 180, 2, 2, 0.1);
INSERT INTO order_items VALUES (293, 181, 14, 4, 0.0);
INSERT INTO order_items VALUES (294, 182, 7, 1, 0.05);
INSERT INTO order_items VALUES (295, 183, 8, 4, 0.2);
INSERT INTO order_items VALUES (296, 183, 6, 2, 0.0);
INSERT INTO order_items VALUES (297, 184, 6, 2, 0.0);
INSERT INTO order_items VALUES (298, 184, 3, 4, 0.0);
INSERT INTO order_items VALUES (299, 185, 12, 3, 0.1);
INSERT INTO order_items VALUES (300, 185, 11, 3, 0.05);
INSERT INTO order_items VALUES (301, 186, 4, 2, 0.0);
INSERT INTO order_items VALUES (302, 187, 6, 2, 0.05);
INSERT INTO order_items VALUES (303, 188, 7, 1, 0.0);
INSERT INTO order_items VALUES (304, 189, 7, 1, 0.0);
INSERT INTO order_items VALUES (305, 190, 4, 1, 0.0);
INSERT INTO order_items VALUES (306, 190, 12, 4, 0.0);
INSERT INTO order_items VALUES (307, 190, 15, 1, 0.05);
INSERT INTO order_items VALUES (308, 191, 1, 1, 0.0);
INSERT INTO order_items VALUES (309, 191, 4, 1, 0.0);
INSERT INTO order_items VALUES (310, 192, 3, 2, 0.2);
INSERT INTO order_items VALUES (311, 193, 7, 2, 0.0);
INSERT INTO order_items VALUES (312, 194, 3, 3, 0.0);
INSERT INTO order_items VALUES (313, 195, 7, 2, 0.1);
INSERT INTO order_items VALUES (314, 195, 12, 3, 0.05);
INSERT INTO order_items VALUES (315, 195, 6, 2, 0.05);
INSERT INTO order_items VALUES (316, 196, 8, 1, 0.0);
INSERT INTO order_items VALUES (317, 197, 5, 1, 0.0);
INSERT INTO order_items VALUES (318, 198, 9, 2, 0.05);
INSERT INTO order_items VALUES (319, 199, 14, 3, 0.0);
INSERT INTO order_items VALUES (320, 200, 7, 2, 0.2);
INSERT INTO order_items VALUES (321, 201, 11, 3, 0.0);
INSERT INTO order_items VALUES (322, 201, 10, 4, 0.05);
INSERT INTO order_items VALUES (323, 202, 11, 2, 0.0);
INSERT INTO order_items VALUES (324, 203, 6, 2, 0.0);
INSERT INTO order_items VALUES (325, 204, 4, 1, 0.0);
INSERT INTO order_items VALUES (326, 205, 7, 2, 0.05);
INSERT INTO order_items VALUES (327, 206, 4, 2, 0.05);
INSERT INTO order_items VALUES (328, 206, 6, 1, 0.0);
INSERT INTO order_items VALUES (329, 207, 15, 3, 0.1);
INSERT INTO order_items VALUES (330, 207, 9, 1, 0.0);
INSERT INTO order_items VALUES (331, 208, 6, 1, 0.0);
INSERT INTO order_items VALUES (332, 209, 4, 1, 0.0);
INSERT INTO order_items VALUES (333, 210, 9, 2, 0.05);
INSERT INTO order_items VALUES (334, 210, 5, 1, 0.0);
INSERT INTO order_items VALUES (335, 210, 14, 3, 0.0);
INSERT INTO order_items VALUES (336, 211, 5, 1, 0.0);
INSERT INTO order_items VALUES (337, 211, 14, 4, 0.0);
INSERT INTO order_items VALUES (338, 211, 4, 1, 0.2);
INSERT INTO order_items VALUES (339, 212, 7, 1, 0.0);
INSERT INTO order_items VALUES (340, 212, 11, 4, 0.0);
INSERT INTO order_items VALUES (341, 212, 13, 3, 0.1);
INSERT INTO order_items VALUES (342, 213, 14, 4, 0.0);
INSERT INTO order_items VALUES (343, 213, 15, 3, 0.0);
INSERT INTO order_items VALUES (344, 213, 5, 1, 0.0);
INSERT INTO order_items VALUES (345, 214, 13, 1, 0.1);
INSERT INTO order_items VALUES (346, 214, 7, 1, 0.05);
INSERT INTO order_items VALUES (347, 215, 5, 1, 0.0);
INSERT INTO order_items VALUES (348, 215, 14, 3, 0.0);
INSERT INTO order_items VALUES (349, 215, 15, 2, 0.0);
INSERT INTO order_items VALUES (350, 216, 15, 3, 0.0);
INSERT INTO order_items VALUES (351, 217, 8, 1, 0.05);
INSERT INTO order_items VALUES (352, 217, 10, 2, 0.0);
INSERT INTO order_items VALUES (353, 218, 3, 1, 0.1);
INSERT INTO order_items VALUES (354, 218, 8, 1, 0.05);
INSERT INTO order_items VALUES (355, 219, 9, 2, 0.0);
INSERT INTO order_items VALUES (356, 219, 15, 3, 0.0);
INSERT INTO order_items VALUES (357, 220, 3, 3, 0.0);
INSERT INTO order_items VALUES (358, 221, 2, 2, 0.05);
INSERT INTO order_items VALUES (359, 221, 9, 2, 0.0);
INSERT INTO order_items VALUES (360, 222, 5, 1, 0.1);
INSERT INTO order_items VALUES (361, 223, 14, 2, 0.0);
INSERT INTO order_items VALUES (362, 223, 1, 2, 0.0);
INSERT INTO order_items VALUES (363, 224, 7, 2, 0.0);
INSERT INTO order_items VALUES (364, 225, 15, 3, 0.0);
INSERT INTO order_items VALUES (365, 226, 7, 2, 0.0);
INSERT INTO order_items VALUES (366, 227, 13, 4, 0.0);
INSERT INTO order_items VALUES (367, 228, 1, 1, 0.0);
INSERT INTO order_items VALUES (368, 229, 12, 4, 0.1);
INSERT INTO order_items VALUES (369, 229, 9, 1, 0.05);
INSERT INTO order_items VALUES (370, 230, 9, 2, 0.2);
INSERT INTO order_items VALUES (371, 231, 9, 1, 0.0);
INSERT INTO order_items VALUES (372, 231, 3, 2, 0.0);
```

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INSERT INTO order_items VALUES (373, 232, 15, 4, 0.1);
INSERT INTO order_items VALUES (374, 233, 14, 3, 0.05);
INSERT INTO order_items VALUES (375, 233, 13, 4, 0.2);
INSERT INTO order_items VALUES (376, 234, 13, 2, 0.2);
INSERT INTO order_items VALUES (377, 235, 12, 4, 0.0);
INSERT INTO order_items VALUES (378, 235, 1, 2, 0.2);
INSERT INTO order_items VALUES (379, 236, 9, 2, 0.0);
INSERT INTO order_items VALUES (380, 237, 5, 1, 0.0);
INSERT INTO order_items VALUES (381, 238, 1, 2, 0.0);
INSERT INTO order_items VALUES (382, 239, 13, 1, 0.0);
INSERT INTO order_items VALUES (383, 239, 2, 1, 0.0);
INSERT INTO order_items VALUES (384, 240, 7, 1, 0.0);

-- PART 3: ANALYSIS QUERIES

-- Q1. Business summary (KPIs)
SELECT COUNT(DISTINCT o.order_id)    AS total_orders,
       COUNT(DISTINCT o.customer_id) AS active_customers,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount)), 2) AS total_revenue,
       ROUND(SUM(oi.quantity * (p.price * (1 - oi.discount) - p.cost)), 2) AS total_profit,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount))
             / COUNT(DISTINCT o.order_id), 2) AS avg_order_value
FROM orders o
JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p     ON p.product_id = oi.product_id;

-- Q2. Monthly revenue, growth and running total
WITH monthly AS (
    SELECT strftime('%Y-%m', o.order_date) AS month,
           SUM(oi.quantity * p.price * (1 - oi.discount)) AS revenue
    FROM orders o
    JOIN order_items oi ON oi.order_id = o.order_id
    JOIN products p     ON p.product_id = oi.product_id
    GROUP BY month
)
SELECT month,
       ROUND(revenue, 2) AS revenue,
       ROUND((revenue - LAG(revenue) OVER (ORDER BY month)) * 100.0
             / LAG(revenue) OVER (ORDER BY month), 1) AS growth_pct,
       ROUND(SUM(revenue) OVER (ORDER BY month), 2) AS running_total
FROM monthly
ORDER BY month;

-- Q3. Top 5 products by revenue
SELECT p.name AS product,
       p.category,
       SUM(oi.quantity) AS units_sold,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount)), 2) AS revenue
FROM order_items oi
JOIN products p ON p.product_id = oi.product_id
GROUP BY p.product_id
ORDER BY revenue DESC
LIMIT 5;

-- Q4. Revenue, profit and margin by category
SELECT p.category,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount)), 2) AS revenue,
       ROUND(SUM(oi.quantity * (p.price * (1 - oi.discount) - p.cost)), 2) AS profit,
       ROUND(100.0 * SUM(oi.quantity * (p.price * (1 - oi.discount) - p.cost))
             / SUM(oi.quantity * p.price * (1 - oi.discount)), 1) AS margin_pct
FROM order_items oi
JOIN products p ON p.product_id = oi.product_id
GROUP BY p.category
ORDER BY revenue DESC;

-- Q5. Cities ranked by revenue
SELECT c.city,
       COUNT(DISTINCT o.order_id) AS orders,
       ROUND(SUM(oi.quantity * p.price * (1 - oi.discount)), 2) AS revenue,
       RANK() OVER (ORDER BY SUM(oi.quantity * p.price * (1 - oi.discount)) DESC) AS city_rank
FROM customers c
JOIN orders o       ON o.customer_id = c.customer_id
JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p     ON p.product_id = oi.product_id
GROUP BY c.city
ORDER BY city_rank;

-- Q6. Top 10 customers and their share of revenue
WITH customer_sales AS (
    SELECT c.customer_id, c.name, c.city,
           SUM(oi.quantity * p.price * (1 - oi.discount)) AS revenue
    FROM customers c
    JOIN orders o       ON o.customer_id = c.customer_id
    JOIN order_items oi ON oi.order_id = o.order_id
    JOIN products p     ON p.product_id = oi.product_id
    GROUP BY c.customer_id
)
SELECT name, city,
```

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       ROUND(revenue, 2) AS revenue,
       ROUND(100.0 * revenue / (SELECT SUM(revenue) FROM customer_sales), 1) AS share_pct
FROM customer_sales
ORDER BY revenue DESC
LIMIT 10;

-- Q7. Customers who never placed an order
SELECT c.customer_id, c.name, c.city, c.signup_date
FROM customers c
LEFT JOIN orders o ON o.customer_id = c.customer_id
WHERE o.order_id IS NULL;
```